

Roberto Villafuerte

AI Engineer | Quito, Ecuador

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I design and build AI and data systems, with an emphasis on the architecture that makes them reliable, secure, and maintainable.

Portfoliovillafortech.com

LinkedInlinkedin.com/in/robertovillafuerte

GitHubgithub.com/VillaforTech

EDUCATION

- 01

BEng, Computer Science Engineering
Universidad San Francisco de Quito (USFQ)
2022 - Expected December 2026

Scholarship recipient. Final semester with capstone project in progress.
- 02

Exchange studies, Data Science
University of Helsinki
August - December 2025

Average: 4.83/5.0. MSc-level courses: Engineering of ML Systems, Big Data Platforms, Trustworthy ML, and Software Architectures.

LEADERSHIP & TEACHING

- Coordinator**

AI Safety Ecuador2025
- Competitive Programming Instructor**

Competitive Programming Club2025

CREDENTIALS

- Technical AI Safety**

BlueDot ImpactJanuary 2026
- Machine Learning Specialization**

IBMOctober 2025
- 2nd Place, IEEEExtreme 18.0, Ecuador**

IEEEOctober 2024

LANGUAGES

- English**

Native or bilingual proficiency
- Spanish**

Native or bilingual proficiency
- Chinese**

HSK 2

SELECTED SKILLS

Data ArchitectureMachine Learning EngineeringData EngineeringSoftware ArchitectureDatabase DesignMachine LearningGenerative AIMLOps

PostgreSQLDatabase SecurityBackend DevelopmentCloud ComputingPythonSQLDocker

EXPERIENCE

- 01

Applied AI Engineer
COMPUMAX

- Translate product and business requirements into architecture, data flows, backend services, and implementation plans.
- Develop AI and data product capabilities with reliability, security, maintainability, and scalability in view.

August 2026 - Present
- 02

Lead Data Architect
SaliHub

- Designed PostgreSQL structures and migrations for a multi-tenant health platform, including access controls, consent, auditability, and reporting.
- Added database health checks, initial tests, and source-aware deduplication across devices; documented technical decisions.

February 2026 - June 2026
- 03

AI Engineer
COMPUMAX

- Built a support-automation system with FastAPI, Chatwoot, Qdrant, PostgreSQL/pgvector, Redis, and Docker Compose.
- Implemented retrieval, intent routing, human escalation, logging, and internal adapters, with a human-only fallback.

December 2025 - February 2026
- 04

Machine Learning Engineer Intern
Datalysis Group

- Developed Python and Mage pipelines for data preparation, feature engineering, model evaluation, prediction, alerts, and export.
- Integrated PostgreSQL ingestion and export, kept predictions separate from evaluation metrics, and corrected schemas, write policies, and documentation.

February 2025 - June 2025

SELECTED PROJECTS

- Auditable ML policy lab**
Academic project · University of Helsinki

Classification lab with separate training, validation, and test stages; fairness reports, traceable artifacts, and an API with explicit contracts.

[Case study and code](#)
- Bio-inspired optimization benchmark**
Open-source project · Python

Reproducible comparison of GA, PSO, ACO, and GWO on test functions, routes, and schedules, with configurable seeds, saved metrics, and convergence curves.

[Code on GitHub](#)